BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: 8197 / 8196

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Objective) Group - I

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8197

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Two non-parallel lines intersect each other at:

(A) 1 point
(B) 0 point
(C) ∞ point
(D) 2 points

Q2. Equation of a straight line passing through P(c , d) and parallel to x-axis is:

(A) x = 0
(B) y = 0
(C) x = d
(D) y = d

Q3. Normal form of equation of straight line is:

(A) y = mx + c
(B) x sin(90° - α) + y cos(90° - α) = p
(C) x/a + y/b = 1
(D) x = y/2 - 5/2

Q4. ax + b > 0 is:

(A) An identity
(B) A linear equation
(C) Equation
(D) Inequality

Q5. For hyperbola b^2 = ?

(A) c2 - a2
(B) a2 - c2
(C) c2 + a2
(D) ac - 1

Q6. Parametric equations of a circle are:

(A) $x = a\cos\theta , y = b\sin\theta$
(B) $x = a\sin\theta , y = b\cos\theta$
(C) $x = a\cos\theta , y = a\sin\theta$
(D) $x = b\cos\theta , y = a\sin\theta$

Q7. The equation $ax^2 + by^2 + 2gx + 2fy + c = 0$ will represent circle if:

(A) $a < b$
(B) $a = b$
(C) $a > b$
(D) $a \neq b$

Q8. If terminal point B of vector AB coincides with its initial point A, then such a vector is called:

(A) Null vector
(B) Unit vector
(C) Coincident vector
(D) Free vector

Q9. If α, β, γ are direction angles of a vector, then:

(A) $0 < \alpha < \pi/2$
(B) $0 \leq \alpha \leq \pi/2$
(C) $0 < \alpha < \pi$
(D) $0 \leq \alpha \leq \pi$

Q10. If $\bar{u} = a\bar{i} + b\bar{j} + c\bar{k}$, then projection of $\bar{u}$ along $\bar{k}$ is equal to:

(A) a
(B) b
(C) c
(D) $\bar{u} \cdot \bar{k}$

Q11. The equations of the form $x = a\cos\theta , y = a\sin\theta$ are called:

(A) Implicit equations
(B) Explicit equations
(C) Parametric equations
(D) Homogeneous equations

Q12. Domain of $f(x) = 2 + \sqrt{(x - 1)}\ \forall x \in \mathbb{R}$ is:

(A) $[-1 , +\infty)$
(B) $[0 , +\infty)$
(C) $[1 , +\infty)$
(D) $[2 , +\infty)$

Q13. If $f(x) = c^3$, where c is any constant, then f'(x) = ?

(A) 3c2
(B) c2
(C) 3/c
(D) 0

Q14. If $y = x^4 + 2x^2 + 3$, then dy/dx = ?

(A) $4x\sqrt{(y - 1)}$
(B) $4x\sqrt{(y - 2)}$
(C) $4x\sqrt{(y - 3)}$
(D) $4x\sqrt{(y - 4)}$

Q15. At a point of maximum value of a function, its derivative is:

(A) Zero
(B) Positive
(C) Negative
(D) Infinite

Q16. If $y = \sin 3x$, then $y_2 = ?$

(A) 3cos 3x
(B) -9 sin 3x
(C) -27 cos 3x
(D) 81 sin 3x

Q17. $\int_0^{\sqrt{3}} dx/(1 + x^2) = ?$

(A) π/6
(B) π/4
(C) π/3
(D) π/2

Q18. $\int (\sin x + \cos x)/\sqrt{(1 + \sin 2x)}\, dx = ?$ is :

(A) x + c
(B) sin x + c
(C) cos x + c
(D) $\cos^2 x + c$

Q19. $\int \tan^2 x\, dx = ?$

(A) tan x + x + c
(B) 2 tan x sec2 x + c
(C) sec x - x + c
(D) tan x - x + c

Q20. $\int \ln x\, dx = ?$

(A) xlnx + c
(B) xlnx - x + c
(C) x/lnx + x + c
(D) lnx + x + c

Mathematics (Subjective) Group - I

Time Allowed: 02:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 8197

Note: SECTION - I

Q2. (Marks 16) QUESTION NO. 2 Attempt any EIGHT parts:

i Show that parametric equations $x = a\cos\theta$, $y = b\sin\theta$ represent the equation of Ellipse $x^2/a^2 + y^2/b^2 = 1$

ii If $f(x) = \sqrt{(x + 1)}$, $g(x) = 1/x^2$, find (fog)(x)

iii Evaluate the limit: Lim $x \to 0$ (sin ax)/(sin bx)

iv Discuss the continuity of $f(x) = \{ 2x + 5 , x \leq 2 ; 4x + 1 , x > 2 \}$ at $x = 2$

v Use definition to find the derivative of x(x - 3) w.r.t 'x'

vi Differentiate $x^4 + 2x3 + x^2$ w.r.t 'x'

vii Differentiate $(1 + x2)^n$ w.r.t. x^2

viii Find dy/dx when x = y sin y

ix If $y = e^{-2x} \sin 2x$, find dy/dx

x Find dy/dx when y = sinh-1(x3)

xi Use Maclaurin Series to prove that $ex = 1 + x + x^2/2! + x^3/3! + x^4/4! + \ldots$

xii Find the interval where $f(x) = 4 - x^2$, $x \in (-2 , 2)$ is increasing or decreasing in the given domain.

Q3. (Marks 16) QUESTION NO. 3 Attempt any EIGHT parts:

i Use differentials, find dy/dx and dx/dy of $x^2 + 2y^2 = 16$

ii Evaluate $\int \sin^2 x \, dx$

iii Find $\int dx/[x(\ln 2x)^2]$

iv Evaluate $\int \sin^{-1} x \, dx$

v Evaluate $\int_1^2 \ln x \, dx$

vi Find area above the x-axis, bounded by curve $y^2 = 3 - x$ from x = -1 to x = 2

vii Solve differential equation $1 + \cos x \tan y \, dy/dx = 0$

viii Find point three-fifth of way along the line segment from A(-5 , 8) to B(5 , 3)

ix Two points P and O' are given in xy-coordinate system. Find XY-coordinates of P. P(3/2 , 5/2) ; O'(-1/2 , 7/2)

x Find an equation of line through (-4 , -6) and perpendicular to the line having slope -3/2

xi Express the system $3x + 4y - 7 = 0$, $2x - 5y + 8 = 0$, $x + y - 3 = 0$ in matrix form and check whether three lines are concurrent.

xii Find lines represented by $x^2 - 2xy \sec\alpha + y^2 = 0$

Q4. (Marks 18) QUESTION NO. 4 Attempt any NINE parts:

i Graph the solution set of linear inequality $5x - 4y \leq 20$ in xy-plane.

ii Define corner point of solution region.

iii Find center and radius of the circle $5x^2 + 5y^2 + 14x + 12y - 10 = 0$

iv Find equation of parabola whose focus is F(-3 , 4) and directrix is 3x - 4y + 5 = 0

v Find length of the tangent drawn from the point (-5 , 4) to the circle $5x^2 + 5y^2 - 10x + 15y - 131 = 0$

vi Find focus and vertices of Ellipse $x^2/9 + y^2/4 = 1$

vii Find equation of tangent to conic y2 = 4ax at (at2 , 2at)

viii Find equation of hyperbola with center (0 , 0), focus (6 , 0) vertex (4 , 0).

ix If O is origin and OP = AB, find the point P when A and B are (-3 , 7) and (1 , 0) respectively.

x Find direction cosines of vector v = i - j - k

xi Find cosine of the angle θ between vectors u = 3i + j - k , v = 2i - j + k

xii A force F = 7i + 4j - 3k is applied at P(1 , -2 , 3), find its moment about Q(2 , 1 , 1)

xiii Find the volume of the parallelepiped determined by u = i + 2j - k , v = i - 2j + 3k , w = i - 7j - 4k

> Note: SECTION - II Attempt any THREE questions. Each question carries 10 marks.

Q5.

Q6.

Q7.

Q8.

Q9.

GROUP – II

Mathematics (Objective) Group - II

Time Allowed: 30 Minutes	**Maximum Marks:** 20	**Paper Code:** 8196

> Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The suitable substitution for $\sqrt{(x^2 - a^2)}$ to be integrated:

(A) x = a sinθ

(B) x = a secθ

(C) x = a tanθ

(D) x = a cosθ

Q2. $\int (ax + b)^n \, dx$ = :

(A) (ax + b)^{n+1}/[a(n + 1)] + c

(B) (ax + b)^{n+1}/[b(n + 1)] + c

(C) (ax + b)^{n+1}/[a(n - 1)] + c

(D) a(ax + b)^{n+1}/(n + 1) + c

Q3. $\int \sqrt{(1 - \cos 2x)}\, dx$ = :

(A) $-\sqrt{2} \cos x + c$

(B) $\sqrt{2} \sin x + c$

(C) $\sqrt{2} \cos x + c$

(D) $-\sqrt{2} \sin x + c$

Q4. $\int e^x (1/x + \ln x)\, dx$ = :

(A) $(1/x)\, ex + c$

(B) $ex(\ln x) + c$

(C) $e^x/\ln x + c$

(D) $\ln x/e^x + c$

Q5. $d/dx\ (y^n)$ = :

(A) ny^{n-1}

(B) ny^{n+1}

(C) $ny^{n-1}\, dy/dx$

(D) $ny^{n-1}\, dx/dy$

Q6. $d/dx\ (3^x)$ = :

(A) $3x \ln 3$

(B) $3x$

(C) $x3^{x-1}$

(D) 3^{x+1}

Q7. If $f(x) = 1/(x - 1)$, then $f'(2)$ = :

(A) -1

(B) 1

(C) 0

(D) 2

Q8. $f(x) = -3x^2$ has maximum value at:

(A) x = -2

(B) x = -1

(C) x = 0

(D) x = 1

Q9. The function $f(x) = (x + 2)^2$ is:

(A) Even

(B) Odd

(C) Both A and B

(D) Neither even nor odd

Q10. $\lim_{x \to 0} (1 + 3x)^{2/x}$ = :

(A) e2

(B) e8

(C) e6

(D) e4

Q11. $(\bar{i} \times \bar{k}) \times \bar{i}$ = :

(A) 1

(B) $-\bar{j}$

(C) 0

(D) $\bar{i}$

Q12. $|\cos\alpha\, \bar{i} + \sin\alpha\, \bar{j} + 0\bar{k}|$ = :

(A) 0

(B) 1

(C) 2

(D) -1

Q13. If $\bar{a} + \bar{b} + \bar{c} = 0$ then:

(A) $\bar{a} \times \bar{b} \times \bar{c} = 0$ (B) $\bar{a} \times \bar{b} = \bar{b} \times \bar{c} = \bar{c} \times \bar{a}$

(C) $\bar{a} \cdot \bar{b} = \bar{b} \cdot \bar{c} = \bar{c} \cdot \bar{a}$ (D) $\bar{a} = \bar{b} = \bar{c}$

Q14. Focus of the parabola $x^2 = -16y$ is:

(A) (0 , 4) (B) (0 , -4)

(C) (4 , 0) (D) (-4 , 0)

Q15. A circle is called a point circle if:

(A) r = 1 (B) r = 0

(C) r = 2 (D) r = 3

Q16. Eccentricity of ellipse is:

(A) e = 0 (B) e > 1

(C) 0 < e < 1 (D) e = 1

Q17. The point (-1 , 2) satisfies the inequality:

(A) x - y > 4 (B) x - y ≥ 4

(C) x + y < 4 (D) x + y > 5

Q18. Equation of horizontal line through (7 , -9) is:

(A) y = -9 (B) y = 7

(C) x = -9 (D) x = 7

Q19. If m_1 and m_2 are the slopes of two lines then lines are perpendicular if:

(A) $m_1m_2 = 0$ (B) $m_1m_2 + 1 = 0$

(C) $m_1m_2 + 2 = 0$ (D) $m_1 = m_2$

Q20. Distance of point (1 , -2) from y-axis is:

(A) 2 (B) 1

(C) 3 (D) 4

Mathematics (Subjective) Group - II

Time Allowed: 02:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 8196

Note: SECTION - I

Q2. (Marks 16) QUESTION NO. 2 Attempt any EIGHT parts:

i If f(x) = { x + 2 , x ≤ -1 ; c + 2 , x > -1 } , then find c so that Lim x→-1 f(x) exists.

ii Evaluate Lim x→∞ (x/(1 + x))x

iii If g(x) = 3/(x - 1) , x ≠ 1 ; then find gog(x)

iv Determine whether f(x) = 3x/(x² + 1) is even or odd.

v Differentiate (2x - 3)/(2x + 1) w.r.t x

vi Find dy/dx if x = θ + 1/θ and y = θ + 1

vii Differentiate cos√x + √(sin x) w.r.t x

viii Differentiate √(tan x) w.r.t x

ix Find f'(x) if f(x) = ln(e^x + e^-x)

x Find y_2 if $x^3 - y^3 = a^3$

xi Prove that cos x = 1 - x²/⌊2 + x⁴/⌊4 - x⁶/⌊6 + ...

xii Determine the interval in which f(x) = sin x is decreasing ; x ∈ (-π , π)

Q3. (Marks 16) QUESTION NO. 3 Attempt any EIGHT parts:

i Find dy and δy for the function y = √x when x changes from 4 to 4.41

ii Evaluate ∫ (3x² - 2x + 1) dx

iii Evaluate the integral ∫ (1 - x²)/(1 + x²) dx

iv Evaluate ∫ x³lnx dx

v Evaluate ∫ 2x/(x² - a²) dx

vi Solve the definite integral $\int_{-1}^{3}$ (x³ + 3x2) dx

vii Find the area between x-axis and the curve y = cos ½x from x = -π to x = π

viii Find 'h' such that points A(-1 , h) , B(3 , 2) and C(7 , 3) are collinear.

ix Find the slope and inclination of the line joining the points (4 , 6) and (4 , 8).

x Find the equation of line through (-4 , 7) and parallel to the line 2x - 7y + 4 = 0

xi Check whether the lines 4x - 3y - 8 = 0 ; 3x - 4y - 6 = 0 and x - y - 2 = 0 are concurrent or not.

xii Find the angle between the pair of lines x² + 2xy secα + y² = 0

Q4. (Marks 18) QUESTION NO. 4 Attempt any NINE parts:

i Indicate solution set of linear inequalities 3x + 7y ≥ 21 , x - y ≤ 2

ii Define optimal solution.

iii Find center and radius of the circle x² + y² - 6x + 4y + 13 = 0

iv Find length of tangent drawn from point (-5 , 4) to the circle $5x^2 + 5y^2 - 10x + 15y - 131 = 0$

v Find the vertex and directrix of parabola $x^2 = 5y$

vi Find equation of ellipse with data vertices (-1 , 1) , (5 , 1) Foci : (4 , 1) , (0 , 1)

vii Find equation of hyperbola with data Foci (0 , ±9) , directrices y = ±4

viii Find equation of normal to $x^2/a^2 - y^2/b^2 = 1$ at (a secθ , b tanθ)

ix Find unit vector in the direction of vector $v = -\sqrt{3}/2\ \bar{i} - 1/2\ \bar{i}$

x Find direction cosines of vector v = 6i - 2j + k

xi Show that the set of points P(1 , 3 , 2) , Q(4 , 1 , 4) and R(6 , 5 , 5) forms a right triangle.

xii Compute cross product $\bar{b} \times \bar{a}$ if $\bar{a} = 3i - 2j + k$, $\bar{b} = i + j$

xiii Prove that vectors i - 2j + 3k , -2i + 3j - 4k , i - 3j + 5k are coplanar.

> Note: SECTION - II Attempt any THREE questions. Each question carries 10 marks.

Q5.

Q6.

Q7.

Q8.

Q9.